

Mini Project 2: Financial Ratios & DuPont Analysis

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1 Data Preparation and Winsorization

All the firm-level observations were taken from the Compustat Fundamental Annual (*funda*) database from 2018 to 2022. The sample was limited to conventional industrial, domestic, and consolidated firm-year observations for five years. Along with the variables needed for the three-factor DuPont decomposition, I also kept two-digit historical SIC codes, the market-to-book ratio, and other company traits such as liquidity, solvency, and size.

The final sample has 31,360 firm-year observations and 8,106 different firms after the screening conditions were applied. Before doing the main analysis, winsorization was applied because raw accounting data often have extreme values due to very small denominators, strange reporting results, or big temporary losses. Table 1 shows how certain summary statistics change when using 1% and 99% winsorization in SAS.

Panel A is the results directly from the raw data which includes a significant amount of extreme and economically unrealistic values with the highest ROE of 1000.00, the highest Profit Margin of 2033.00, the highest Total Asset Turnover of 514000.00, and the highest Equity Multiplier of 29586.74. In Panel B, which is the outcome after winsorization, these extreme values were capped at both the upper and lower tails, however, the medians still stayed the same. This may indicate that the procedure worked exactly as it planned: it made the outliers less important without taking away any observations or significantly changing the center of the distributions.

Table 1: Selected Summary Statistics Before and After Winsorization

Variable	N	Mean	Median	Min	Max
<i>Panel A: BEFORE Winsorization</i>					
ROE	31,360	-1.4605	0.0570	-1801.05	1000.00
PM	31,360	-14.0708	0.0416	-84221.67	2033.00
TAT	31,360	16.9993	0.4143	0.0000	514000.00
EM	31,360	6.5368	2.3831	1.0000	29586.74
<i>Panel B: AFTER 1% and 99% Winsorization</i>					
ROE	31,360	-0.1966	0.0570	-8.0468	1.3839
PM	31,360	-2.6733	0.0416	-60.8315	1.4461
TAT	31,360	0.5884	0.4143	0.0023	3.0786
EM	31,360	4.2243	2.3831	1.0335	33.9270

After winsorization, Table 2 shows the mean and median values of ROE and the three DuPont parts. The mean values of ROE and Profit Margin are much lower than their medians

which means that the sample is still left-skewed, even after taking those extreme outliers out. By contrast, Total Asset Turnover stays positive and fairly stable, and the Equity Multiplier stays well above 1 on average. This shows that leverage is still an important part of the ROE structure.

Table 2: Mean and Median of ROE and DuPont Components After Winsorization

Variable	Mean	Median
ROE	-0.1966	0.0570
PM	-2.6733	0.0416
TAT	0.5884	0.4143
EM	4.2243	2.3831

2 Trend of DuPont 3 Factors (2018–2022)

The DuPont framework breaks down ROE into three parts: profit margin, total asset turnover, and equity multiplier. Figures 1 to 3 show how the three DuPont components have changed over the past five years, and Figure 4 illustrates how ROE has changed in the same time period while Table 3 shows the average and median values for each year.

Table 3: Yearly Mean and Median of ROE and DuPont Components

Year	ROE Mean	PM Mean	TAT Mean	EM Mean	ROE Median	PM Median	TAT M
2018	-0.1577	-2.1751	0.6443	4.0295	0.0656	0.0475	
2019	-0.2144	-2.3449	0.6005	4.2432	0.0551	0.0394	
2020	-0.2404	-2.8067	0.5357	4.3218	0.0289	0.0243	
2021	-0.1342	-2.9767	0.5498	4.0792	0.0718	0.0536	
2022	-0.2380	-3.0398	0.6130	4.4483	0.0618	0.0425	

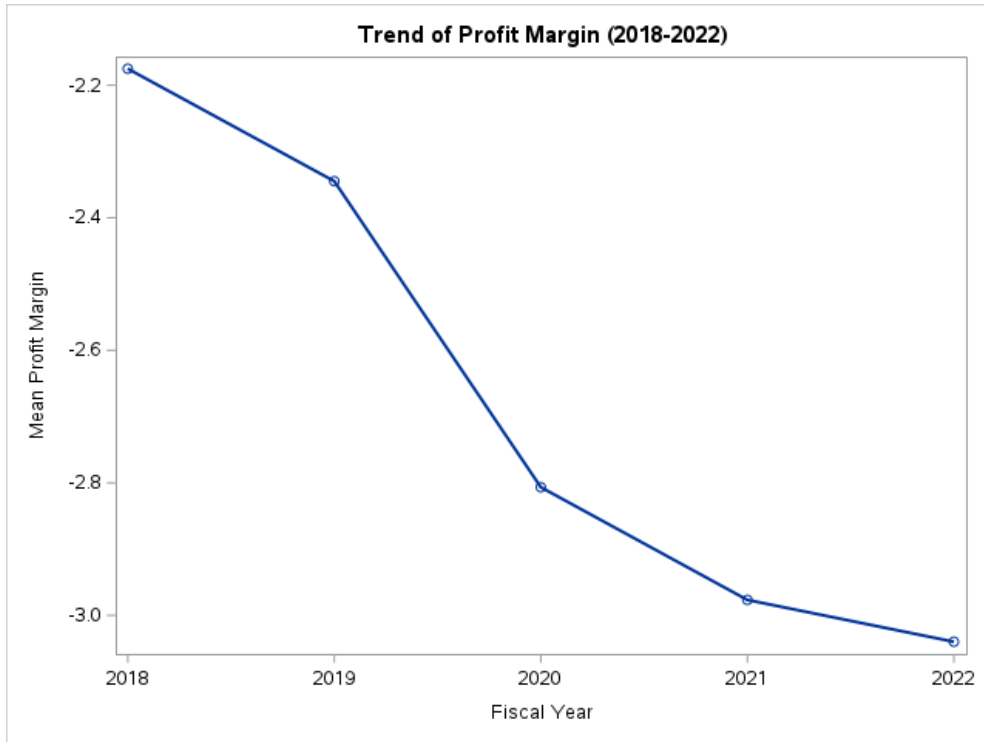


Figure 1: Trend of Profit Margin (2018–2022)

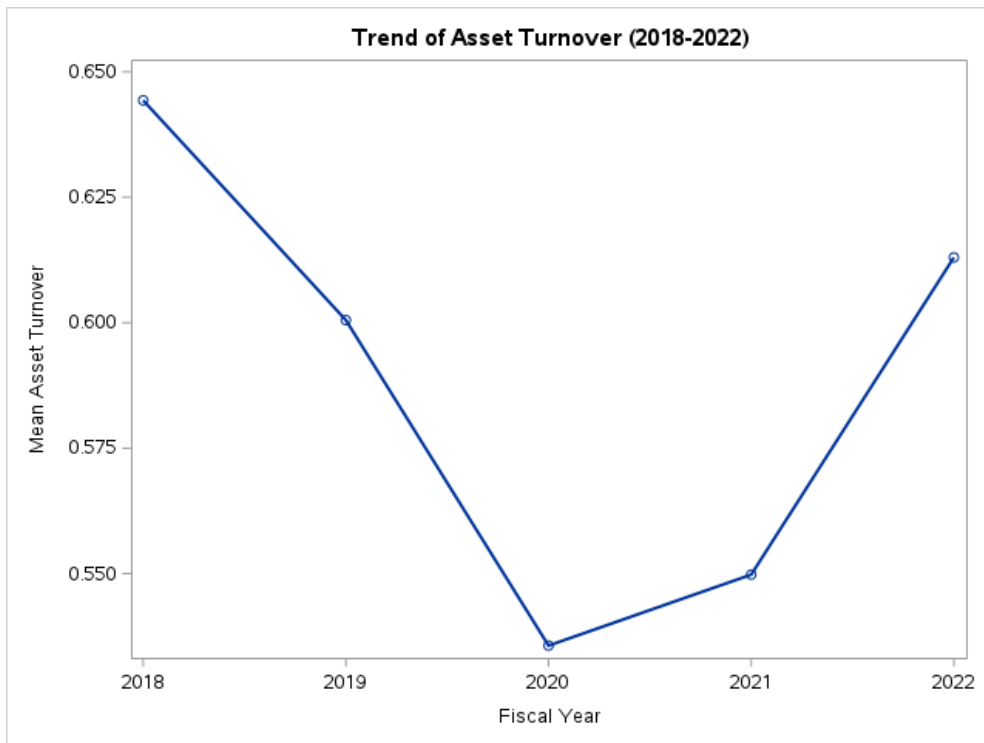


Figure 2: Trend of Asset Turnover (2018–2022)

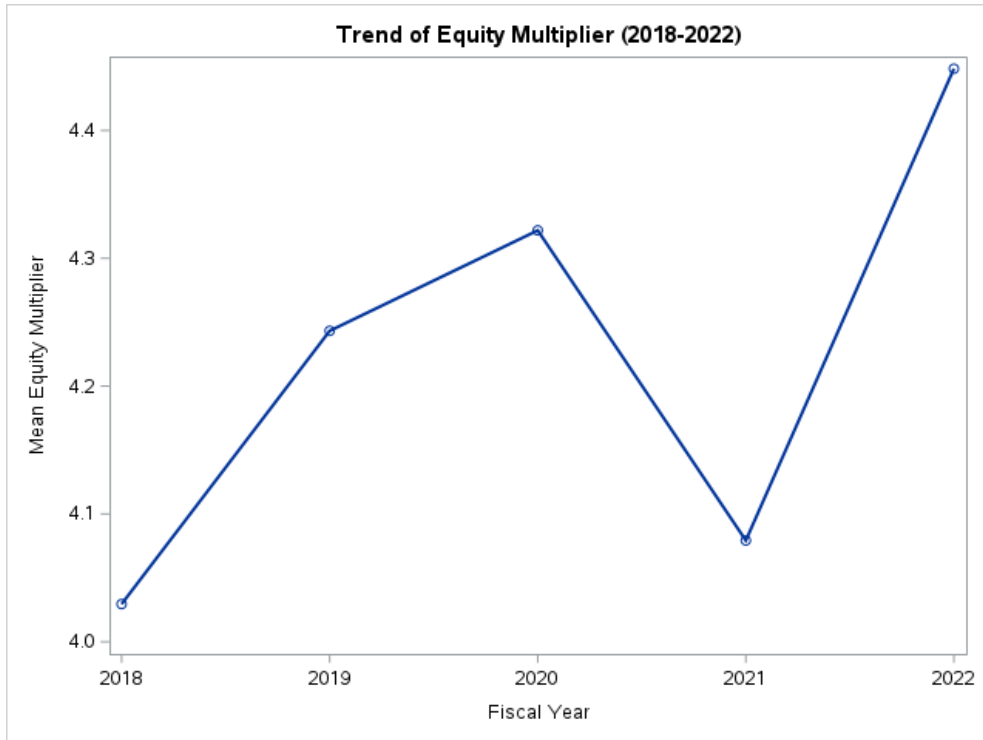


Figure 3: Trend of Equity Multiplier (2018–2022)

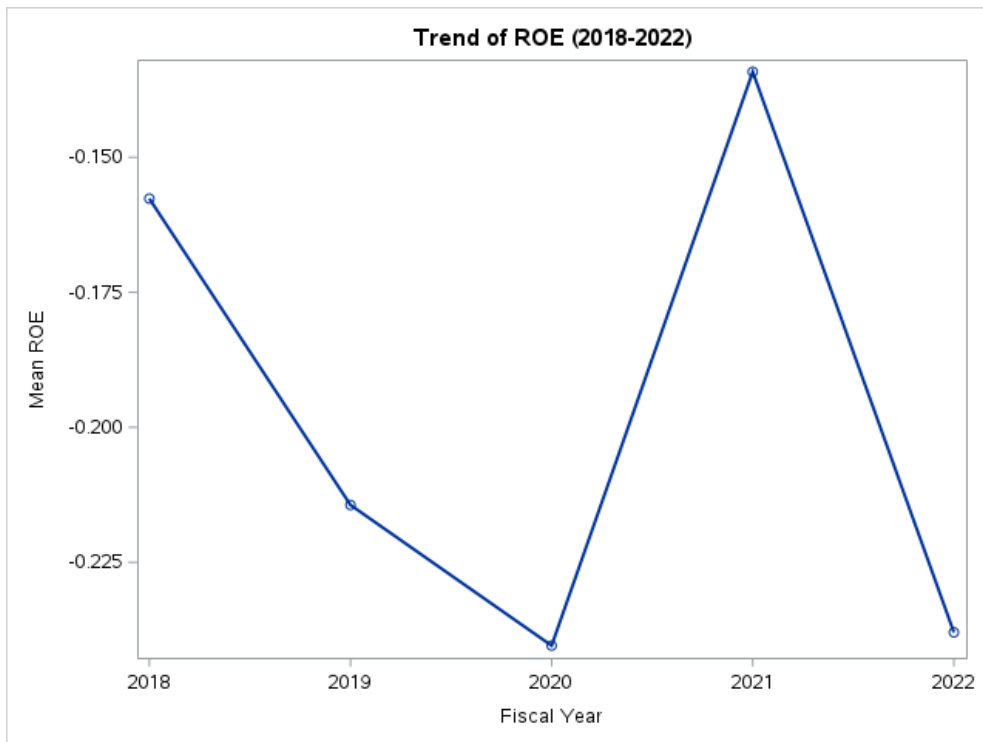


Figure 4: Trend of ROE (2018–2022)

The trend points to a story that beyond figures. Profit Margin gets worse over time and stays weak after 2020, which shows that low profitability is the main reason why average ROE is low. In 2020, Total Asset Turnover drops a lot, which aligns with the pandemic shock happened that year. The ratio partially recovers in 2021 and 2022. Meanwhile, the Equity Multiplier stays

pretty high throughout the samples and goes up even further in 2022. This may suggest that companies have used more leverage while their profits were still under pressure.

The ROE series shows a similar pattern. ROE gets worse from 2018 to 2020, and gets better in 2021, then it gets worse again in 2022. This change fits with the idea that the small rise in sales after 2020 wasn't enough to make up for the fact that profit margins were still weak under the Covid. In other words, the overall drop in ROE during this time seems to be more like a result of low profits due to the market than a permanent drop in operating efficiency. The rise in the Equity Multiplier also suggests that some companies were using more debt to boost returns.

3 Determinants of ROE: Regression Analysis

The analysis also includes an OLS regression of ROE on the three DuPont parts and some company traits, such as the Current Ratio, Debt-to-Equity ratio, Market-to-Book ratio, and company size, to give more proof. The results help show if these relationships are still statistically significant after taking into account other differences between firms.

The model retains strong overall, as shown in Table 4. The F -statistic is 1607.00 and the R^2 is 0.3352. The coefficients for Profit Margin, Total Asset Turnover, and the Equity Multiplier are all positive and statistically significant. This is mostly in line with the DuPont framework where the Current Ratio is also positively related to ROE, which means that in this sample, companies with better short-term liquidity tend to do better. On the other hand, the Debt-to-Equity ratio is quite negative, which means that a higher debt load is linked to lower returns for shareholders when all other factors are held constant. The size of a company is positively related to ROE, but the coefficient on MTB is negative in this case.

Table 4: OLS Regression Results for ROE

Variable	Estimate	Standard Error	t Value	Pr > t
Intercept	-1.5003	0.0276	-54.34	< .0001
Profit Margin (PM)	0.0122	0.0004	30.71	< .0001
Asset Turnover (TAT)	0.2819	0.0105	26.84	< .0001
Equity Multiplier (EM)	0.1192	0.0151	7.90	< .0001
Current Ratio (CR)	0.0337	0.0016	21.36	< .0001
Debt to Equity (DE)	-0.2140	0.0154	-13.85	< .0001
Market to Book (MTB)	-0.0073	0.0008	-9.45	< .0001
Size	0.1659	0.0027	62.35	< .0001
<i>Model Fit: F-Value = 1607.00, R-Square = 0.3352, p-value < 0.0001</i>				

4 Conclusion

This small project uses SAS to clean, winsorize, and look at a big sample of Compustat firm-years using the DuPont framework. The main finding is that the low average ROE from 2018 to 2022 is mainly due to low profits probably resulting from the Covid's impacts. Asset turnover goes down in 2020 but then climbs back up. Leverage stays high and goes back up by 2022. The regression results back up the same general conclusion by showing that profitability, turnover,

and leverage are still important factors that affect ROE even when liquidity, solvency, market value, and size are taken into account.

In short, it looks like the drop in shareholder returns during this time was more of a problem with making money than with being efficient. Higher leverage may have helped some companies deal with the pressure from the market, but it still wasn't enough to bring average ROE back to where it was before.